Kiera McCormick — CV

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RESEARCH INTERESTS

My research focuses on large language models for scientific and real-world understanding, including retrieval-augmented generation, prompt engineering, and question-answering systems. I am particularly interested in human–AI comparison and cooperation on natural language processing tasks, evaluation of human–computer interaction in high-stakes or scientific settings, and benchmarking LLM behavior. More broadly, my interests include open-source science, interdisciplinary applications of machine learning, machine translation, and AI safety.

EDUCATION

Johns Hopkins University — **Baltimore, MD**

Master of Science in Computer Science — Expected Graduation: May 2027

- **Concentration:** Human Language Technology
- **Relevant Coursework:** NLP for Computational Social Science, AI-Enabled Software Engineering, Machine Social Intelligence, Natural Language Processing, Human Language Technology

Loyola University Maryland — **Baltimore, MD**

Bachelor of Science in Engineering — Graduated May 2025

- **Concentration:** Electrical and Computer Engineering
- **Minor:** Mathematics
- **GPA:** 3.745
- **Dean's List Honors:** Fall 2021, Spring 2022, Fall 2022, Spring 2023, Fall 2023, Spring 2024, Fall 2024, Spring 2025
- **Relevant Coursework:** Physics I & II, Linear Circuit Analysis, Statics, Calculus I, II, III, Electronics, Digital Logic, Quantum Computing, Signals and Systems, Ordinary Differential Equations, Programming Tools, Object-Oriented Engineering Design, Probability and Statistics, FPGA Design, Communications, Linear Algebra, Electromagnetics, Microprocessor-Based Systems, Advanced Linear Algebra, Quantum Computing II, Power Systems, Engineering Systems Analysis

Danish Institute for Study Abroad **Copenhagen, DK**

Core Course: Holocaust and Genocide *Spring 2024*

RESEARCH AND WORK EXPERIENCE

JHU | Human Language Technology Center of Excellence **Baltimore, MD**
Research Intern, Summer Camp for Applied Language Exploration (SCALE)
June 2026 – Present

- Contributed to a summer research program focused on evaluation methods for multimodal retrieval-augmented generation systems.
- Developed a novel evaluation metric to better capture how well AI systems tailor information to different user needs, improving on limitations of existing approaches.
- Validated the metric through human-agreement studies and empirical testing across multiple real-world event datasets.

Johns Hopkins University **Baltimore, MD**
Research Assistant
January 2026 – Present

- Conducting research at the intersection of AI and forensic science in collaboration with NIST.
- Surveying AI tools across forensics-adjacent domains, building structured analyses of AI adoption patterns, and contributing to stakeholder documentation on responsible AI integration in forensic laboratories.

Center for Astrophysics | Harvard & Smithsonian **Cambridge, MA**
AstroAI Summer Intern
January 2025 – August 2025

- Engineered prompts to generate concise and discriminative summaries about astrophysical X-ray sources from Chandra Archives data and created visualizations to understand the prompting effects.
- Extracted embedded physical parameters to evaluate LLMs' understanding of scientific knowledge.
- Applied sparse autoencoders to reveal novel patterns and relationships among astrophysical objects.

Space Telescope Science Institute and Johns Hopkins University **Baltimore, MD**
Summer Astronomy Space Program and the Annual Jelinek Memorial Summer Workshop
May 2024 – August 2024

- Examined the application of artificial intelligence and machine learning, particularly Large Language Models (LLMs), in astronomy research. Assessed the potential of LLMs to enhance research efficiency and reliability.
- Designed and implemented LLM-powered chatbots for deployment on Slack, specializing in prompt engineering, conducting comparisons between open and closed-

source models, integrating information retrieval pipelines, and creating evaluation benchmark datasets.

Omega Technical Services at Los Alamos National Laboratory **Los Alamos, NM**
Engineering Intern
May 2023 – August 2023

- Collaborated with the industrial engineering team at the Los Alamos National Laboratory, focusing on task scheduling optimization through databases and automation. Transformed a previously manual task of job scheduling to an automated system.
- Utilized database coding and conducted data analysis using tools such as MicroStrategy, Excel, SQL, Python, VBA, and Confluence, contributing to the creation of metrics and dashboards for effective data visualization.
- Worked towards automating an extract, transform, load (ETL) pipeline through MariaDB, MS Task Scheduler, and Alteryx.

Loyola University Maryland **Baltimore, MD**
Physics Teaching Assistant and Tutor
September 2022 – May 2025

- Supported undergraduate instruction in Physics I and II through problem-solving sessions and individualized tutoring, emphasizing quantitative reasoning and conceptual understanding.

University of Delaware **Newark, DE**
Junior Physics Intern
March 2020 – August 2020

- Contributed to an astrophysical research project analyzing the spectral characteristics and luminosity of massive stars. Developed Python scripts using Jupyter Notebook to process and visualize complex stellar data.
- Created data visualizations of spectral lines and magnitude measurements, enhancing the interpretation of stellar properties.

PUBLICATIONS

- *Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries*
    - **McCormick, K.,** Martínez-Galarza, J. R., 2025, *NeurIPS: ML4PS*, accepted.
- *From Queries to Criteria: Understanding How Astronomers Evaluate LLMs*
    - **McCormick, K.**, Hyk, A., Zhong, M., Ciucă, I., et al., 2025, *COLM*, accepted.
- *Designing an Evaluation Framework for Large Language Models in Astronomy Research*
    - Wu, J. F., Hyk, A., **McCormick, K.,** Ye, C., et al., 2024, arXiv preprint, arXiv:2405.20389.

- *pathfinder: A Semantic Framework for Literature Review and Knowledge Discovery in Astronomy*
    - Iyer, K. G., Yunus, M., O'Neill, C., Ye, C., Hyk, A., **McCormick, K.,** et al., 2024, *ApJS 275 38,* arXiv:2408.01556.

CONFERENCES

Machine Learning conference/workshop paper reviewer for *NeurIPS* for the EvalEval Workshop in 2024.

Volunteer at the *AstroAI Workshop*

Gender Minorities and Women in Physics Summit **Johns Hopkins University**

Poster Presentation September 14, 2024

- Discussed work on the application of Large Language Models in Physics and Astronomy.

American Astronomical Society 245 **National Harbor, Maryland**

iPosters January 11-16, 2025

- Presented "Evaluating Large Language Models in Astronomy Research" at the Machine Learning Methods session.
- Co-authored "pathfinder: a Semantic Framework for Literature Review and Knowledge Discovery in Astronomy" at the Machine Learning Methods session.

Talks

- Co-authored "Evaluating LLM Tools: Insights from Astronomer Interactions with a RAG-Based Slack Chatbot" at the Technology Developments in Outreach, Education, and Research session.

AstroAI Workshop **Cambridge, Massachusetts**

Poster Presentation July 7-11, 2025

- Presented "Evaluating Large Language Models in Astronomy Research".

Conference on Language Modeling **Québec, Canada**

Poster Presentation October 7-10, 2025

- Presented "From Queries to Criteria: How Astronomers Evaluate LLMs".

Conference on Neural Information Processing Systems **San Diego, California**

Poster Presentation December 2-7, 2025

- Presented “Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries” at the Machine Learning for the Physical Sciences Workshop.

STScI Language AI in the Space Sciences **Baltimore, Maryland**

Poster Presentation March 9-12, 2026

- Presented “Encoding and Understanding Astrophysical Information in Large Language Model-Generated Summaries”.

American Society for Crime Laboratory Directors **Grand Rapids, Michigan**

Booth Staff

May 18-21, 2026

- Staffed the National Institute of Standards and Technology booth, gathering practitioner perspectives on AI adoption in forensic science.

TECHNICAL & RESEARCH SKILLS

- **Programming**: Python, SQL/SQLite, Java, C++, Git/GitLab, HTML
- **ML/AI**: familiarity with PyTorch and TensorFlow, LLMs, retrieval-augmented generation and information retrieval, representation learning, model evaluation and benchmarking, prompt engineering
- **Scientific/Engineering tools**: MATLAB (certified), Simulink, LT Spice, Active HDL, MicroStrategy
- **Hardware/Systems**: Raspberry Pi, Arduino, FPGA, embedded systems

Completion of the 2024 Summer School on Human Language Technology at Johns Hopkins University

SELECTED PROJECTS

A2CPS Discovery Agent **January 2026 – May 2026**

- Co-developed a natural language interface for querying the A2CPS biomedical dataset, translating plain-English questions into read-only SQL.
- Contributed to a retrieval-augmented generation pipeline for surfacing relevant study documentation alongside query results.
- Implemented features in a React/Hono application using the Anthropic Claude API.

Regional Variation in ChatGPT Use Across U.S. States **January 2026 – May 2026**

- Applied BERTopic to 68,668 first-turn prompts from the WildChat corpus to study geographic and socioeconomic variation in ChatGPT use across 36 U.S. states.
- Found that education and income correlate with specialized creative uses, while generic image-generation requests are more common in lower-education and lower-income states.

Testing Social Reasoning in LLMs **January 2026 – May 2026**

- Designed and evaluated a multi-dimensional benchmark of 40 original scenarios spanning Theory of Mind, Pragmatics, and Chain-of-Thought reasoning across 12 state-of-the-art LLMs.
- Recruited a human baseline (n=12) to quantify the gap between model and human-level social reasoning.

Hotspot Homing Robot **September 2024 – May 2025**

- Developed an autonomous mobile robot designed to detect and localize rogue access points in office environments, sponsored and mentored by members of the Applied Signal Technology sector at Raytheon.
- Integrated wireless signal processing and detection techniques with navigation systems, such as Simultaneous Localization and Mapping, to create a comprehensive security solution.
- Collaborated with an interdisciplinary team to design a user-friendly interface for real-time threat visualization and reporting.

EXTRACURRICULAR ACTIVITIES

Johns Hopkins Women's Club Lacrosse Team **Baltimore, MD**
Academic All-American September 2025 – Present
Society of Women Engineers **Baltimore, MD**
President February 2023 – May 2025
Engineering Industrial Advisory Board **Baltimore, MD**
Class of 2025 Representative December 2023 – May 2025
Loyola University Maryland Honors Program **Baltimore, MD**
Member and Peer Mentor September 2021 – May 2025
Haig Scholar **Baltimore, MD**
Member February 2024 – May 2025
Loyola Women's Club Lacrosse Team **Baltimore, MD**
Academic All-American September 2022 – May 2025
Institute of Electrical and Electronics Engineers **Baltimore, MD**
Member October 2021 – May 2025